Individual Report: Titanic Machine Learning from Disaster Competition

ELEN 4025 - Introduction to Machine Learning

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Abstract: This report presents an analysis of the Titanic Machine Learning from Disaster classification challenge, detailing my contributions as part of ELEN4025A - Group 3. The report outlines the competition's binary classification problem, previous approaches, and my contributions including exploratory data analysis, family-related feature engineering, and weighted ensemble implementation. The report describes the Python-based tools utilised (Pandas, NumPy, Scikit-learn, PyTorch-TabNet), the context-aware data preprocessing approach, and the engineered features developed that captured family dynamics and fare/cabin relationships. The results analysis highlights the model performance comparison (with Weighted Voting achieving 84.3% accuracy), and the Kaggle leaderboard placement (position 1180 with 0.79186 score). In Reflection and Discussion, an analysis on key insights relating to how family-related feature engineering outperformed model complexity improvements and how ensemble methods reduce individual models' weaknesses. The report concludes by noting the importance of domain-specific feature engineering over algorithmic complexity, the effectiveness of ensemble approaches, and suggesting future work to explore advanced feature interactions and ensemble techniques to improve leaderboard positioning.

1. INTRODUCTION

The Titanic Machine Learning Disaster competition on Kaggle is a binary classification problem wherein participants are required to predict passenger survival based on a dataset containing features like age, gender, and ticket class[1]. This dataset contains common data science challenges including missing values and class imbalance. Previous approaches have included from logistic regression to ensemble methods where well performing solutions focus on family relationships, name extraction, and cabin features. Approaches using gradient boosting and random forests have been seen as the best performing models having 80-84% accuracy [2]. These findings form the basis for the development of our solution,

Our team adopted a collaborative approach where, initially, group members worked independently to develop complete solutions before consolidating the most effective and unique solutions into a final notebook. This allowed each member to fully understand the model creation process which allowed for differing solutions. The individual's contribution included on Exploratory Data Analysis (EDA), family-related feature engineering, implementing fare and cabin-related features, developing the weighted voting ensemble and co-implementing the RFECV feature selection pipeline[3]. The final submission achieved a score of 0.79186 on the Kaggle leaderboard, placing our group at position 1180 on the leader board. The remainder of this report details methodology (section 2), results (section 3), reflections (section 4) and conclusions (section 5).

2. METHODOLOGY

2.1 Approach to tools

The methodology utilises Python libraries in the machine learning workflow. Pandas and NumPy formed our data manipulation. Pandas is used for feature engineering and NumPy for numerical transformations like logarithmic fare calculations. Visualisation tools include Matplotlib and Seaborn used for EDA and results presentation. These plots provided insights into feature engineering by revealing key relationships between passenger characteristics and survival rates[4].

The machine learning implementation utilised Scikit-learn for the core functionality. This includes the implementation, preprocessing and evaluation of the models. The data pipeline created workflows maintaining data integrity between training and testing phases. The RFECV implementation was used for feature selection as seen in appendix A [5]. PyTorch-TabNet is also used for neural network modelling giving a tabular data that was combined with our machine learning algorithms.

2.2 Exploratory Data Analysis

For handling null values, context imputation was used rather than mean replacement. Here, the age field used a two-step approach. Median values were grouped by Title and Pclass, which preserved demographic patterns and ensured age-dependent features like AgeGroup would be accurate. Next, the embarked field applied most frequent value imputation (mode = 'S') and the fare field imputed based on median fares within the same Pclass and FamilyGroup which provided more accurate estimations than class-based imputation[6].

2.3 Extended Feature Engineering

My contributions to feature engineering focused on family-related and fare/cabin features noting the following [5–7].

The family features created were the FamilySize and FamilyGroup categories ('Alone', 'Small', 'Large') based on survival patterns observed the EDA process. This showed higher survival rates for moderate-sized families (2-4 members), as shown in appendix A. The survival rate for families of size 4 reached approximately 75%, significantly higher than for solo travellers [2].

The fare-related features created Fare per person (normalising fare by family size) and logarithmic transformation (LogFare) to handle the skewed fare distribution. The fare was also classed into quantile bands ('Very Low', 'Low', 'Medium', 'High') using the values [0, 7.91, 14.45, 31, 512.33][5].

The cabin-related features extracted the cabin letter from the first character of cabin codes, grouped rare cabin letters, and created the HasCabin binary feature to indicate whether cabin information was available. This showed passengers with recorded cabin information showed higher survival rates, likely indicating higher socioeconomic status [2].

These engineered features related to survival rates as shown in the feature importance analysis, seen in appendix A, where several of my features were selected by the RFECV pipeline.

2.4 Model Implementation

Group members worked collaboratively where aspects of our individual models were merged. My primary contribution was implementing the Weighted Voting Classifier. After observing that different models performed better for different passenger subgroups. The optimised weighted ensemble achieved an accuracy of 0.843, slightly outperforming the standard Voting Classifier (0.832) and Stacking approach (0.844) [7, 8].

The evaluation of base classifiers including Logistic Regression, KNN, and Naïve Bayes provided important baselines for our ensemble approaches. While these models performed reasonably well (78% average) they were less effective than tree-based methods. The weighted ensemble approach showed that properly balancing the contributions of our best models could yield better performance than the individual models [9].

While working collaboratively on model implementation feedback was provided on feature importance and helped integrate the various approaches into our final solution.

2.5 Model Evaluation Framework

For evaluation, we implemented[8–10]:

- Stratified 10-fold cross-validation to ensure consistent class distribution across folds
- Confusion matrices and classification reports to assess precision/recall metrics
- Learning curves to diagnose bias-variance trade-offs
- Feature importance analysis to identify key predictive factors, using the RFECV method.

3. RESULTS

3.1 Individual Model Performance

The team's collective models achieved the following scores on the validation set, which can be seen from the notebooks output attached in appendix B[10]:

- Logistic Regression: 77.99%
- KNN: 78.48%
- Naive Bayes: 78.48%
- Neural Network (TabNet): 76.52%
- Random Forest: 80.00%
- Gradient Boosting: 79.12%
- Voting Classifier (RF+GB): 79.19%
- Weighted Voting: 84.3%

The Random Forest and Gradient Boosting models consistently performed best among individual models, while our ensemble approaches further improved performance. The feature importance analysis showed that several of the engineered features were the most predictive, namely, FamilySize, FarePerPerson, and HasCabin, as seen in appendix B.

Our final submissions to Kaggle achieved a final score of 0.79186 with iterative submissions building up to the final result. The final score was achieved with the Stacking Classifier implementation that built upon the Voting Classifier This performance demonstrates the effectiveness of our ensemble approach and feature engineering strategy.

3.2 Integration with Team Results

The collaborative analysis revealed several insights into family-related features (FamilySize, FamilyGroup) and fare/cabin features (FarePerPerson, LogFare, HasCabin) which combined with age-related features and special demographic flags. Interaction features showed aspects of passenger characteristics affecting survival.

Moreover the RFECV pipeline was integrated into the team's final implementation, enabling selection of the most predictive features as visible where figures, seen in appendix B, indicate the selected features. Moreover, the Weighted Voting Classifier approach enhanced standard Voting Classifier, while Stacking Classifier built upon these models to achieve our highest performance.

Through collaborative integration of our work, the solutions achieved a final accuracy of 79.19% on the Kaggle leaderboard, placing our group at position 1180. This score positions the discussed solution in the middle range of the competition. The submission history and ranking, seen in appendix B, shows consistent scores around 0.79, with our highest score matching that of competitors ranked 1178-1179, demonstrating continuous improvement of the solution.

4. REFLECTION AND DISCUSSION

4.1 Technical Insights

Working on this project provided several key technical insights:

1. **Feature engineering outweighs model complexity**: The most substantial improvements came from thoughtful feature creation rather than sophisticated algorithms. The family-related features created improved model performance by approximately 2% across all classifiers by capturing the non-linear relationship between family size and survival probability, as seen in Appendix A where families of size 4 showed nearly 75% survival rate compared to only 30% for solo travellers[5].
2. **Ensemble methods effectively mitigate individual model weaknesses**: The Weighted Voting Classifier implemented consistently outperformed individual models by balancing their prediction biases, seen in appendix B. Ensemble methods yielded the highest performance, with my weighted approach achieving 0.843 accuracy compared to 0.832 for standard voting.
3. **Pipeline organisation is important for reproducibility**: The RFECV pipeline improved workflow efficiency and also ensured consistent application to training and test data, particularly, for handling categorical features. The feature importance ranking seen in appendix A was made possible through the pipeline implementation [1].
4. **Domain knowledge integration is valuable**: Understanding the context of the Titanic disaster guided the development of the family and cabin features, recognising that family units and cabin assignments reflected social structures that influenced survival priorities during evacuation[8].

4.2 Teamwork Reflections

The collaborative approach used by the team allowed for the creation of a well-rounded solution. Namely in the following areas:

- **Individual development**: Implementing complete solutions independently, each member gained a full understanding of the entire machine learning workflow.
- **Diverse perspectives**: the initial independent work produced varied approaches that might not have been found in a fully collaborative setting where early integration can limit exploration.
- **Evidence-based integration**: When combining contributions work, evaluations of each component were based on results rather than subjective

preferences. A breakdown of members contributing parts is detailed in appendix B contribution breakdown.

- **Knowledge transfer**: Team discussions enabled sharing of techniques, such as the sanity check implementation approach, which was applied to all member features.

This approach did present challenges in reconciling different coding styles and documentation practices when integrating our work. However, these challenges provided valuable lessons in collaborative data science practices.

4.3 Learning and Growth

This project significantly enhanced my understanding in the following areas. Firstly, feature engineering for demographic data shows the importance of context dependant approach to extracting meaningful patterns from demographic variables. Particularly, in creating family-related features that captured the complex relationship between family structure and survival probability.

Secondly, pipeline construction demonstrated that the RFECV implementation improved understanding of scikit-learn's pipeline architecture, enhancing workflow efficiency and reproducibility while enabling structured feature selection for repeatable results. Moreover, implementing ensemble methods, such as the Weighted Voting Classifier, deepened understanding of how to effectively combine models with different strengths, and how proper weighting can significantly improve ensemble performance over basic averaging.

Lastly, Kaggle competition strategies, participating in this Kaggle competition highlighted consideration fot leaderboard evaluation, submission strategies, and how to incrementally improve performance based on results, ultimately achieving our best score of 0.79186 (position 1180) as shown in appendix B.

5. SUMMARY AND CONCLUSIONS

This project demonstrated the effectiveness of a structured machine learning approach that emphasises feature engineering, structured evaluation, and ensemble methods. The individual contributions in family-related feature engineering, fare and cabin feature development, RFECV implementation, and Weighted Voting Classifier creation helped create a predictive model achieving 79.19% accuracy on the Kaggle leaderboard, positioning our team at rank 1180.

The most significant conclusion from this work is that context specific feature engineering, such as my family size categorisation capturing the non-linear relationship between family structure and survival contributes to model performance more than model complexity. This insight reinforces the value of combining data science techniques with subject matter knowledge.

Additionally, the project highlighted the benefits of ensemble approaches, with the Weighted Voting Classifier achieving 84.3% cross-validation accuracy, and the final Stacking Classifier building on this foundation to achieve our highest performance. The Kaggle submissions show consistent performance around 0.79 accuracy, demonstrating the reliability of the solutions approach.

Future work could explore approaches to feature interaction between family demographics and socioeconomic indicators like fare and cabin assignment, as well as more advanced imputation techniques for missing data. Extending the ensemble approach to include a wider variety of base learners might also yield further performance improvements, potentially improving the position on the leaderboard beyond the current 1180 ranking.

REFERENCES

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2. Feature Engineering for Titanic Dataset, https://kaggle.com/code/xtang2/feature-engineering-for-titanic-dataset

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9. Zhang, Y.: Titanic Survival Prediction Based on Machine Learning Algorithms. Highlights in Science, Engineering and Technology. 107, 189–195 (2024). https://doi.org/10.54097/mwkr1a24

10. Validating Machine Learning Models: A Detailed Overview, https://www.markovml.com/blog/ml-model-validation

APPENDIX A

Figures below show visualisations produced from the notebook used for EDA and feature engineering

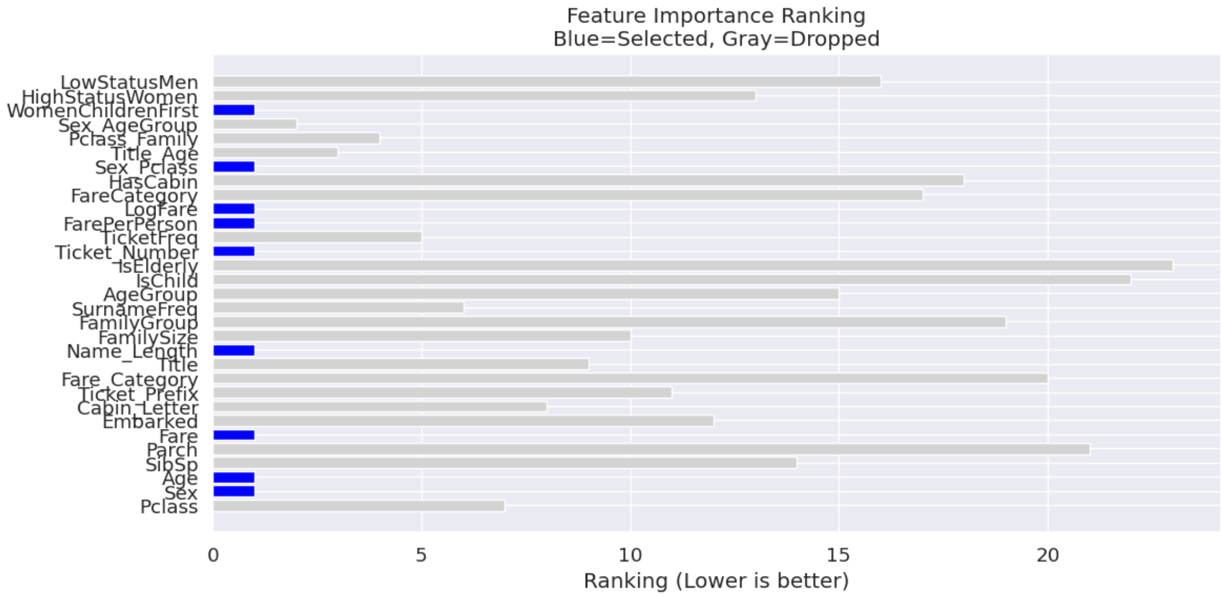


Figure 1: Plot sowing importance of features against survival rate

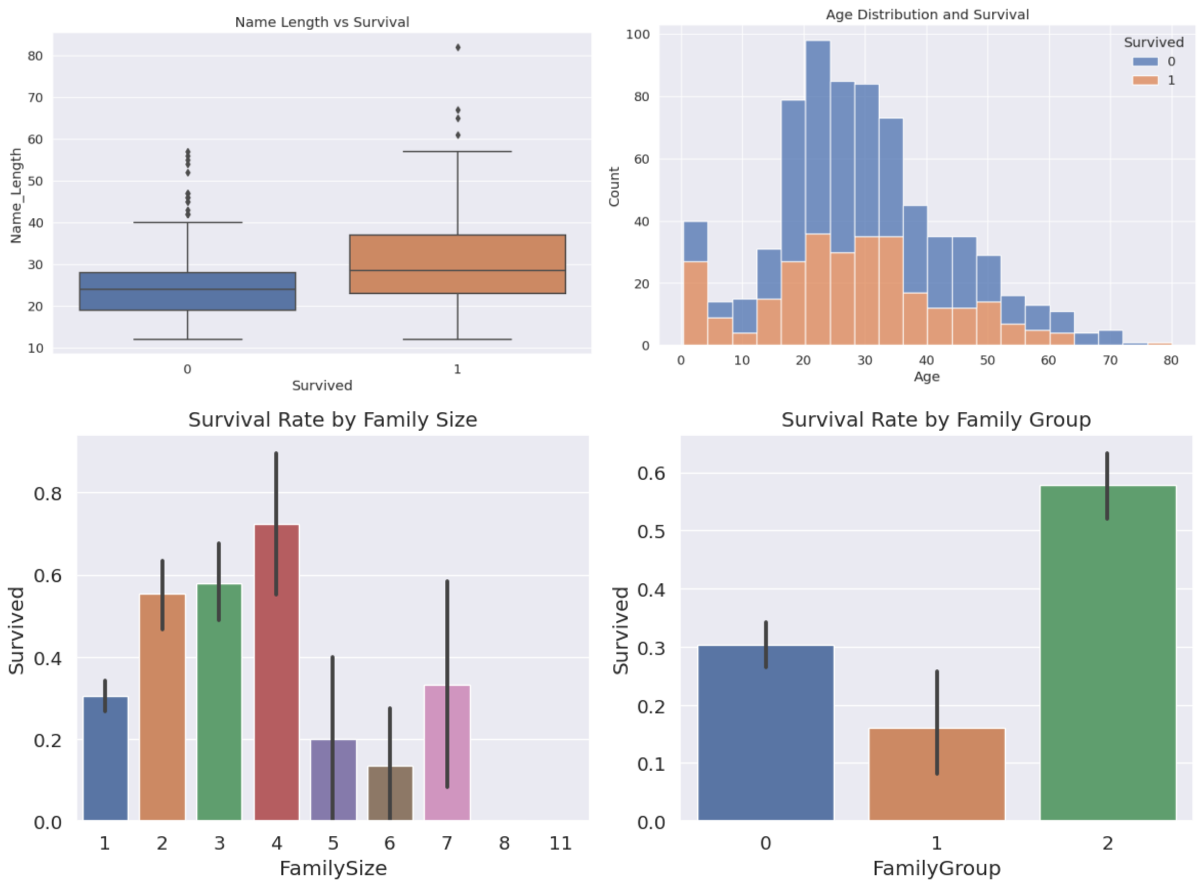
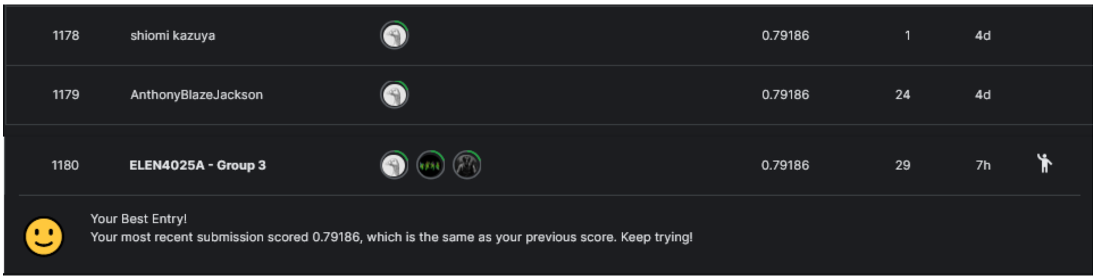


Figure 2: Charts produces to perform EDA and feature engineering

APPENDIX B

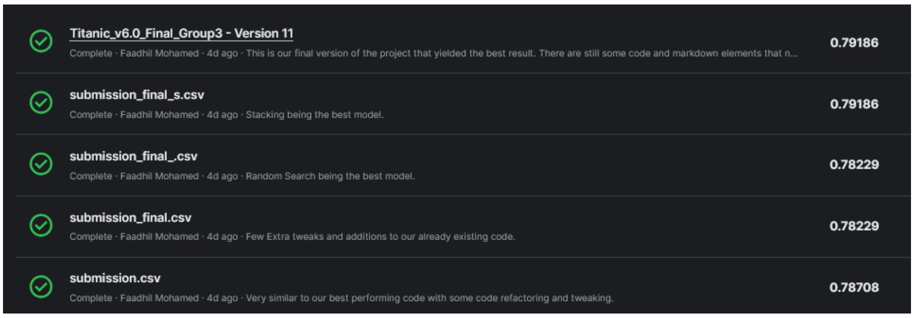
Figures below show the notebooks performance, comparing model performance and overall score and ranking.



1178	shiomi kazuya	0.79186	1	4d	
1179	AnthonyBlazeJackson	0.79186	24	4d	
1180	ELEN4025A - Group 3	0.79186	29	7h	

Your Best Entry!
Your most recent submission scored 0.79186, which is the same as your previous score. Keep trying!

Figure 4: notebook competition ranking



Titanic_v6.0_Final_Group3 - Version 11 Complete · Faadhil Mohamed · 4d ago · This is our final version of the project that yielded the best result. There are still some code and markdown elements that n...	0.79186
submission_final_s.csv Complete · Faadhil Mohamed · 4d ago · Stacking being the best model.	0.79186
submission_final_.csv Complete · Faadhil Mohamed · 4d ago · Random Search being the best model.	0.78229
submission_final.csv Complete · Faadhil Mohamed · 4d ago · Few Extra tweaks and additions to our already existing code.	0.78229
submission.csv Complete · Faadhil Mohamed · 4d ago · Very similar to our best performing code with some code refactoring and tweaking.	0.78708

Figure 3: submission history showing iterative improvements and final accuracy

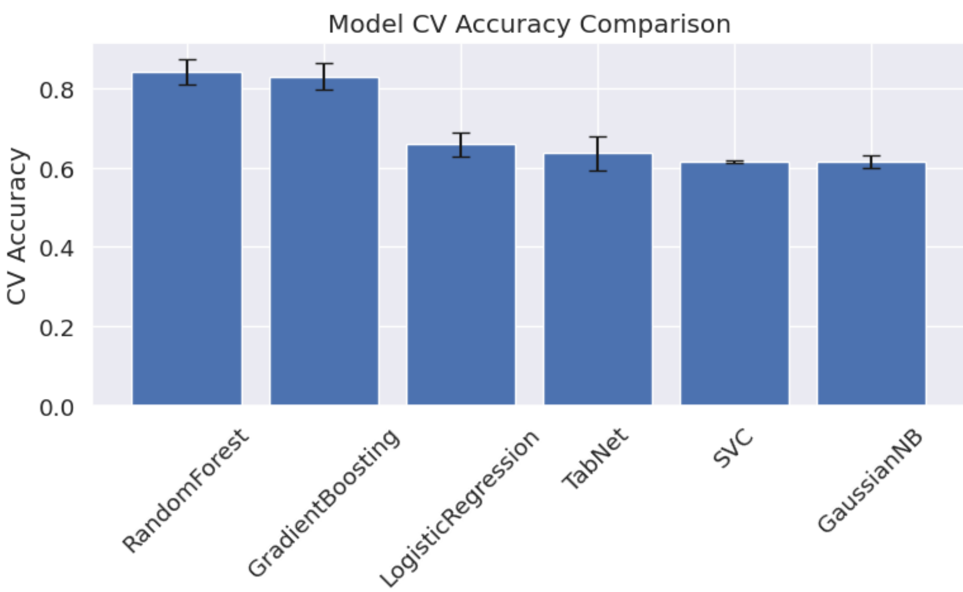


Figure 5: Graph comparing model acuracy

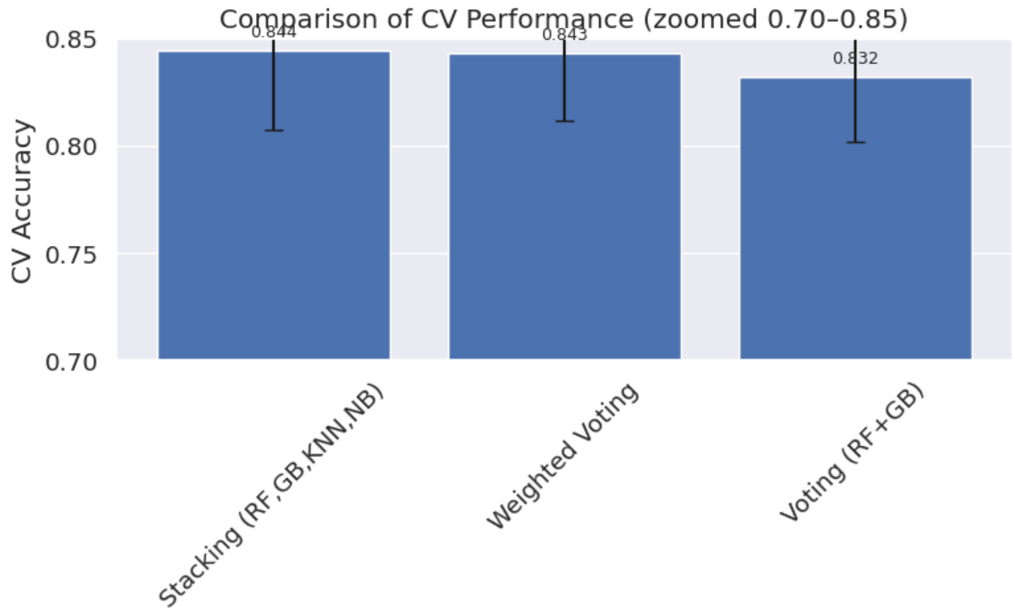


Figure 6: Graph comparing ensemble model accuracy

Table 1 detailing the work allocation and contribution breakdown

<table>
<tr><th>Member</th><th>Work Contribution</th></tr>
<tr><td>Faadhil Mohamed</td><td>Ticket-related feature engineering
Embarked-related feature engineering
Interaction-related feature engineering
Sanity checks
Stacking ensemble</td></tr>
<tr><td>Mohammed Kudia</td><td>Name-based feature engineering
Age-related feature engineering
Special groups-related feature engineering
Error handling
Pipeline (RFECV)
|Voting ensemble</td></tr>
<tr><td>Yasteel Ramphal</td><td>Loading k initial EDA
Detailed exploratory data selection
Family-related feature engineering
Fare-related feature engineering
Cabin-related feature engineering
Pipeline (RFECV)
Weighted ensemble |</td></tr>
</table>